Erratum to ISF Papers VI, VII and VIII: a missing Jacobian in the per-lens perturber count

Every detectable-perturber count in the three papers is high by a factor 1.4, every Poisson significance by $\sqrt{1.4}$; the joint weak-lensing fit, the profiles and the falsification logic stand

Christopher K. Merrill

With computational collaboration from Claude Fable 5.1 (Anthropic)

October 11, 2026

Abstract

Paper VI of the Independent Structure Formation (ISF) series counts the other-branch galaxies detectable per strong-lens sightline by integrating a stellar mass function, written per dex of stellar mass, over the logarithm of halo mass. The conversion between the two carries the Jacobian $d\log M_*/d\log M_{\rm halo}$, which for the paper's single power-law abundance matching is the constant $1/1.4$, and the script omitted it. Every quantity downstream of that integral is therefore high by exactly 1.4: Paper VI's 1.08 detectable perturbers per lens at $N = 6$ is 0.77; Paper VII's calibration constant $C_{\rm sub} = 0.514$ is 0.367 and its per-lens counts at $N = 2$–10 fall from 0.51–1.20 to 0.37–0.86; Paper VII's fiducial Euclid expectation of 4,050 detections is 2,893 and its all-ISF falsification threshold of 1,844 is 1,307; Paper VIII's wide-survey expectation of 4,463–5,955 is 3,188–4,253, its threshold of 2,037–2,733 is 1,444–1,939, its DR1 null exclusion of $\geq 24\sigma$ is $\geq 20\sigma$, and its 312-system in-principle test of 12.6–16.8 expected perturbers at 5.0–5.8σ is 9.0–12.0 at 4.2–4.9σ. Nothing is strengthened and nothing is fatal; the series' falsification statements are restated here with the corrected numbers. One qualitative statement does change: in the ISF-true forecast scenario the weaker Euclid term no longer collapses the posterior onto the $N = 6$ that generated the mock, which splits between $N = 6$ and $N = 8$. The audit also found that Paper VI's text quotes Schechter parameters its script does not use; the published numbers come from the script's parameters, and the text is corrected to them. Paper VII's joint weak-lensing plus substructure fit ($N = 10$, $k_t^{\rm phys} = 0.52\,h\,{\rm Mpc}^{-1}$), Paper VI's convergence profiles and bound-mass table, Paper IX, and Paper X's own counts are untouched. Every number here is printed by a committed script.

1 The error

Paper VI (Merrill, 2025) counts the detectable other-branch galaxies per strong-lens sightline as

$$N_{\rm det} = \int_{10}^{12} (N-1)\big[1-\varepsilon(k_M)\big]\,\Phi\big(M_*(M_{\rm halo})\big) \times V\; d\log_{10} M_{\rm halo}, \quad (1)$$

with Φ the stellar mass function per dex of *stellar* mass, evaluated at the abundance-matched stellar mass $M_* = 10^{10.7}(M_{\rm halo}/10^{12}\,M_\odot)^{1/1.4}$, and the sum taken over $d\log M_{\rm halo}$. Paper VI's equation for $dn_{\rm ISF}/d\log_{10} M$ calls $\Phi_{\rm gal}(M)$ "the galaxy halo mass function (converted from the stellar mass function via the stellar-to-halo mass relation)". That conversion is

$$\frac{dn}{d\log M_{\rm halo}} = \frac{dn}{d\log M_*}\,\frac{d\log M_*}{d\log M_{\rm halo}} = \Phi(M_*)\,\frac{1}{1.4}, \quad (2)$$

and the script that produced the published numbers omits the second factor. Because the matching relation is a single power law the factor is a constant, so every quantity downstream of Eq. 1 is high by exactly 1.4 and every Poisson significance built on it by $\sqrt{1.4} = 1.18$. Reproducing Paper VI's functions with its exact constants gives the

published 1.0806 per lens at $N = 6$, $\varepsilon_\infty = 0.58$ (and 1.80, 2.57 at $\varepsilon_\infty = 0.30$, 0) to three parts in 10^4; with the Jacobian the same code gives 0.772, 1.287, 1.838, a ratio of 1.3996.

The omission was found during the peer review of Paper X (Merrill, 2026c) on 2026-10-10, confirmed the same day against Paper VI's text and script, and declared in Paper X as its concession 6. Paper X gates on Paper VI as published and its own perturber counts use a mapping with unit Jacobian, so they do not change (Section 7).

2 The Schechter parameters

The audit found a second discrepancy that the erratum must settle. Paper VI's text quotes a single Schechter function with $\Phi^* = 0.012$ Mpc^{-3} dex^{-1}, $\log_{10} M_*^{\rm char} = 10.7$ and $\alpha_S = -1.05$, attributed to Baldry et al. (2012). The script that produced the published numbers uses $\Phi^* = 3.96 \times 10^{-3}$, $\log_{10} M_*^{\rm char} = 10.7$, $\alpha_S = -1.16$. Neither is Baldry et al. (2012) as published, which is a double Schechter with $\log_{10} M_*^{\rm char} = 10.66$, $\Phi_1^* = 3.96 \times 10^{-3}$, $\alpha_1 = -0.35$, $\Phi_2^* = 0.79 \times 10^{-3}$, $\alpha_2 = -1.47$. Table 1 gives the per-lens count each set produces. The text's parameters would have given 2.65 per lens, 2.4 times the published value, so the discrepancy is not cosmetic. The correction is to the text: Paper VI's Schechter equation should read $\Phi^* = 3.96 \times 10^{-3}$ Mpc^{-3} dex^{-1}, $\log_{10} M_*^{\rm char} = 10.7$, $\alpha_S = -1.16$, a single-Schechter approximation whose normalisation and characteristic mass follow Baldry et al. (2012) and whose low-mass slope is an intermediate value. The double Schechter as published would give 0.49 per lens with the Jacobian, 37 percent below the corrected 0.77. That is a model change, not an erratum; it is reported here as a check and not adopted, and a future version of the series may take it up.

3 Corrected numbers

Tables 2–4 list every number the three papers quote from the affected chain, published beside corrected. The Paper VII and VIII values come from rerunning the papers' own forecast scripts on the corrected per-lens counts, after each rerun first reproduced the published data product from the published input to a relative 10^{-9}. The corrected products are published beside the originals as `euclid_forecast_v2.npz` and `recalibrated_forecast_v2.npz`; the originals are not altered.

4 What changes qualitatively

Two statements do not survive as written; both concern the ISF-true forecast scenario, in which a mock Euclid count generated at $N = 6$ is added to the current joint fit.

Paper VII, Section 6.1. "$P(N = 6) = 0.92$ (up from 9×10^{-6} currently)" and "the current WL-dominated posterior ($N = 10$ preferred) shifts decisively to $N = 6$" are wrong after the correction. With the smaller count the Euclid term separates $N = 6$ from $N = 8$ by 3.6σ rather than 4.2, which no longer overrides the current fit's preference for $N = 8$–10: the corrected posterior is $P(N = 6) = 0.47$, $P(N = 8) = 0.52$, $P(N = 10) = 3 \times 10^{-4}$, with the joint best fit at $N = 8$. The forecast still excludes $N \leq 4$ and $N = 10$ at the mock; the claim weakens from "Euclid collapses the N posterior" to "Euclid narrows it to $N = 6$–8".

Paper VIII, abstract and Section 4. "Only the full wide survey collapses the posterior onto the truth ($P(N = 6) = 0.97$–1.00)" becomes $P(N = 6) = 0.64$–0.95: the wide survey still prefers $N = 6$, decisively under the flat efficiency model and no longer decisively under the linear and area models. The companion statement, that DR1 falsifies but cannot confirm, survives unchanged in form: the DR1 null exclusion stays above 20σ and the DR1 ISF-true $P(N = 6)$ stays near 10^{-3}.

Everything else is a rescaling. "DR1 is decisive for falsification" holds at $\geq 20\sigma$; "the 312 already-modelled Q1 systems suffice for a first real-data test" holds at $\geq 4.2\sigma$ for $N \geq 6$, which no longer reaches the 5σ the abstract quoted, and at 2.9–3.4σ for all $N \geq 2$.

5 What is unchanged

Verified by reading the code and, where a product exists, by byte comparison after the reruns:

- Paper VII's joint weak-lensing plus substructure fit: the substructure likelihood uses the Meneghetti excess factor $f_{\rm GGSL} = 3[1 + (N -$

Table 1: Detectable perturbers per lens at $N = 6$, $\varepsilon_\infty = 0.58$, under the three Schechter parameter sets, without and with the Jacobian. The published 1.08 is the script set without it; the corrected value is the script set with it.

parameter set	no Jacobian	with Jacobian
text (Φ^* 0.012, α_S −1.05)	2.647	1.891
script (3.96×10^{-3}, −1.16)	**1.081**	**0.772**
Baldry et al. (2012) double Schechter	0.681	0.486

Table 2: Paper VI. $N_{\rm sub}$, the bound-subhalo channel, is 3.2×10^{-4} per lens and stays negligible. The published Euclid row used 1.081 rounded; the corrected row is from the script.

quantity	published	corrected
$N_{\rm det}$ per lens, $\varepsilon_\infty = 0.58$ (abundance table)	1.08	0.77
$N_{\rm det}$ per lens, $\varepsilon_\infty = 0.30$	1.80	1.29
$N_{\rm det}$ per lens, $\varepsilon_\infty = 0.00$	2.57	1.84
ratio to the CDM baseline 0.43	2.5×	1.8×
combined $N_{\rm total}$ per lens (LOS table)	1.13	0.82
combined $f_{\rm LOS}$ / $f_{\rm sub}$	98.5 / 1.5%	98.0 / 2.0%
LOS-fraction shift from ΛCDM	31.5 points	31.0 points
Euclid, $\varepsilon_\infty = 0.58$: per lens / 75,000 / 5%	1.081 / 81,075 / 4,054	0.772 / 57,904 / 2,895
Euclid, $\varepsilon_\infty = 0.30$	1.801 / 135,075 / 6,754	1.287 / 96,506 / 4,825
abstract: "∼4,000 at $> 50\sigma$"	4,054 at 90σ	2,895 at 76σ
2,500 imaging-suitable lenses	2,701	1,930
criterion (1): >1,000 lenses, $\eta = 0.05$	10.4σ	8.8σ

Table 3: Paper VII, Section 6 (75,000 grade-A/B lenses, $\eta = 0.05$, $10^{8.8}\,M_\odot$). Unchanged: $\varepsilon_\infty(N)$, $f_{\rm GGSL}$, the joint fit, and the current $P(N)$.

quantity	published	corrected
$C_{\rm sub}$	0.514	0.367
$N_{\rm det}$ per lens, $N = 2/3/4$	0.51 / 0.77 / 0.93	0.37 / 0.55 / 0.66
$N_{\rm det}$ per lens, $N = 6/8/10$	1.08 / 1.15 / 1.20	0.77 / 0.82 / 0.86
expected detections, $N = 2$ to 10	1,929–4,513	1,378–3,223
fiducial $N = 6$: $n_{\rm exp}$, Poisson σ	4,050, 63.6	2,893, 53.8
fractional precision on $N_{\rm det}$	1.6%	1.9%
ISF-true: $N = 2$ / 3 / 4 from $N = 6$	33.3 / 18.2 / 9.1σ	28.2 / 15.4 / 7.7σ
ISF-true: $N = 8$ / 10 from $N = 6$	4.2 / 7.3σ	3.6 / 6.1σ
ISF-true: $P(N = 6)$, $P(N = 8)$	0.92, 0.08	0.47, 0.52
ISF-true joint best fit	$N = 6$, $k_t = 0.79$	$N = 8$, $k_t = 1.08$
null: $N = 2$ / 6 / 10 excluded at	62 / 90 / 95σ	52 / 76 / 80σ
null: min $\Delta\chi^2$ vs ΛCDM	3,859	2,757
95% threshold, all ISF ($N \geq 2$)	<1,844	<1,307
95% threshold, $N \geq 4$	<3,358	<2,384

Table 4: Paper VIII (measured Q1 demographics; brackets over the flat, linear and area efficiency models). Unchanged: the effective efficiencies 0.050 / 0.041 / 0.038, the stages Q1 497, DR1 14,965, Wide 110,269, and the 312-system effective sightlines 15.6 / 12.7 / 11.7.

quantity	published	corrected
Wide $n_{\rm exp}(N=6)$	4,463–5,955	3,188–4,253
Wide 95% threshold, all ISF	2,037–2,733	1,444–1,939
Wide 95% threshold, $N \geq 4$	3,706–4,966	2,632–3,529
Wide ISF-true $P(N=6)$	0.967–0.999	0.635–0.949
Wide null $\sigma(N=2)$	65.2–75.3	55.1–63.6
DR1 $n_{\rm exp}(N=6)$	606–808	433–577
DR1 95% threshold, all ISF	257–348	180–244
DR1 null $\sigma(N=2)$ / $\sigma(N=6)$	≥ 24.0 / ≥ 34.8	≥ 20.3 / ≥ 29.4
DR1 ISF-true $P(N=6)$	$\sim 10^{-3}$	$\sim 10^{-3}$
DR1 $\sigma(4\,\mathrm{v}\,6)$	3.5–4.1	3.0–3.4
Q1 $n_{\rm exp}(N=6)$	20–27	14–19
312 systems: $n_{\rm exp}(N=6)$	12.6–16.8	9.0–12.0
312 systems, zero found: $N \geq 6$ excluded at	5.0–5.8σ	4.2–4.9σ
312 systems: $N=10$ excluded at	5.3–6.1σ	4.5–5.2σ
312 systems: all $N \geq 2$ excluded at	3.5–4.0σ	2.9–3.4σ

1)(1 − ε_∞)], not Eq. 1; the per-lens count appears in that script only in a printed cross-check. The best fit $N = 10$, $k_t^{\rm app} = 1.32$, $k_t^{\rm phys} = 0.52\,h\,\mathrm{Mpc}^{-1}$, $\Delta\chi^2 = -53.6$, the Limber analysis and the current $P(N)$ stand; `joint_constraint.npz` and `lnL_sub_grid.npz` were not rewritten.

- Paper VI's convergence profiles, the 38 percent convergence excess at the Einstein radius, the bound-mass table, the ε-versus-mass relation, the Meneghetti partial explanation (excess factor, not the integral), the ΛCDM baselines of 0.43 and 0.05 per lens, the sub-galactic contribution below 1 percent, and the scorecard's test outcomes other than the counts. Falsification criteria (2)–(6) are unchanged; criterion (1)'s significance falls from 10.4 to 8.8σ and stays above 5.

- Paper VIII's demographics: the 497 grade-A/B candidates, the 7.88 per square degree rate, the 110,269 wide-survey lenses, the θ_E distribution and the efficiency models.

- Paper IX entirely.

- Paper X (Section 7).

6 Falsification statements, restated

Paper VI, criterion (1). Absence of the 10^{10}–$10^{12}\,M_\odot$ Schechter excess in a sample of more than 1,000 lenses at 5 percent efficiency would falsify ISF at more than 5σ (8.8σ at $N = 6$; was 10.4).

Paper VII. If Euclid substructure counts at $10^{8.8}\,M_\odot$ yield fewer than 1,307 detections across 75,000 grade-A/B strong lenses (was 1,844), the ISF model is excluded at > 95 percent confidence for all $N \geq 2$; fewer than 2,384 (was 3,358) excludes $N \geq 4$.

Paper VIII. If a complete perturber search of the 312 modelled Q1 grade-A/B systems at $10^{8.8}\,M_\odot$ yields zero detections, ISF with $N \geq 6$ is excluded at $\geq 4.2\sigma$ (was $\geq 5\sigma$) and all $N \geq 2$ at $\geq 2.9\sigma$ (was ≥ 3.5); if Euclid DR1 substructure counts fall below 180–244 detections across ∼15,000 grade-A/B lenses (was 257–348), all ISF models ($N \geq 2$) are excluded at > 95 percent confidence; a DR1 null excludes even $N = 2$ at $\geq 20\sigma$ (was ≥ 24).

7 Consistency with Paper X

Paper X replaces Paper VI's intact-NFW perturber with the captured spheroid its own chain derives and maps perturber mass to number with a unit Ja-

cobian. Its corrected per-lens count at $N = 6$ with the extent held, 0.756, sits beside the Jacobian-corrected 0.771 here: at $N = 6$ Paper X's threshold change was this correction in disguise, the profile effect being nil there. Its 312-system expectation for the registered $m = 1$ model with the extent held, 8.8–11.8, sits beside the corrected 9.0–12.0. Paper X's headline numbers are for the $m = 2$–3 family and are not affected; its concession 6 and its Table 3 row "VIII, $N = 6$" (1.080; 16.8 / 13.7 / 12.6; 5.8 / 5.2 / 5.0) now read 0.771; 12.0 / 9.8 / 9.0; 4.9 / 4.4 / 4.2.

8 Reproducibility

The audit and the two reruns are three scripts of a few hundred lines, each gated on reproducing the published product before it prints a corrected number, published with the series' analysis code in the `errata/` folder of the public repository together with the two `_v2` data products. The published Zenodo records of Papers VI (Merrill, 2025), VII (Merrill, 2026a) and VIII (Merrill, 2026b) receive new versions carrying this note; the record numbers the series cites stay valid.

References

Baldry, I. K., et al. 2012, MNRAS, 421, 621

Merrill, C. K. 2025, ISF Paper VI, Galaxy-Mass Dark Substructure from Strong Lensing: A Parameter-Free Test of Independent Structure Formation, Zenodo, doi:10.5281/zenodo.18637044

Merrill, C. K. 2026a, ISF Paper VII: Joint Weak Lensing and Strong Lensing Substructure Constraints on ISF Cross-Branch Correlation Parameters, Zenodo, doi:10.5281/zenodo.18644812

Merrill, C. K. 2026b, ISF Paper VIII: Anchoring the Falsification Forecast to Measured Euclid Q1 Strong-Lens Demographics, Zenodo, doi:10.5281/zenodo.21195841

Merrill, C. K. 2026c, ISF Paper X: The Galaxy-Scale Dark Mass Profile Under Independent Structure Formation, Zenodo, doi:10.5281/zenodo.23286778